Microcontroller-based
Bit Error Rate Test set

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3rd Year Project Interim Report

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I declare that all material described in this report is my own work except where explicitly and individually indicated in the text. This includes ideas described in the text, figures and computer programs.

This report contains _ pages (excluding this page and the appendices) and _ words.

Signed: Vishnu Barath Date: December 4, 2025

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1 Project Description

Communication networks are continously expanding in capacity and speed and being able to ensure signal integrity is paramount. The most common example of such a system is the internet, which uses undersea optical fibres for data transmission. Such networks have become a central component in the modern day world with applications in many important industries, such as telecommucations, cloud services, banking, healthcare and transportation. In accordance with these advancements, techniques to measure the performance of communication channels have evolved. Channels for digital communication as susceptible to incorrect data transmission arising due many possible factors such as noise or interference. A metric of performance for digital communication channels is the Bit Error Rate (BER) which can be characterised via a Bit Error Rate Test (BERT). [1]

The BER of a communication channels is the proportion of bits in a data transmission which have an error. When received data has bit errors it requires either retransmission and/or extra processing from error correction algorithms. Hence, a high BER would have signficant impacts on the channel's datarate capability. [1] Being able to measure this is important in ensuring a channel can transmit data quickly and efficiently. Unfortunately, systems that can measure the BER can be expensive and would be designed for data rates in the Gb/s range. As a result acquiring such equipment could prove difficult and would be excessive since such high data rates aren't always required. [2]

This project aims to construct a system that can accurately measure the BER of a Visible Light Communication (VLC) channel at data rates up to 10Mb/s, while keeping costs relatively low.

2 Literature Review

Building a system to measure a channel's BER can be done via any programmable device such as Microcontrollers or Field Programmable Gate Arrays (FPGA). Low cost BERTs have been developed using such devices. [2, 3, 4] Examples of these are discussed below.

2.1 Serial Communication BER Tester

An FPGA-based device was developed to test the BER of a serial communication channel. It uses a 31-bit Linear Shift Feedback Register (LFSR) to generate a Pseudorandom Binary Sequence (PRBS) which would have a length of $2^{31} - 1$ bits ($\sim 2.14 \times 10^9$). [2, 5]

The system works by transmitting a PRBS to an external channel and receiving the bitstream on the same FPGA. The bits are generated, received and compared simultaneously. It is capable of testing data rates of up to 50Mbps and used the Anritsu MP8302A BERT for reference. [2] By transmitting and receiving on the same FPGA, having to synchronise bitstreams was not an issue. However, this is not good for simulating realistic transmission between two different devices.

2.2 VLC BER Tester

A PCB with logic circuits was designed to test a VLC channel. This is the same type of communication channel this project is meant for. In this case, the bit generator is simply a square wave rather than a PRBS. Effectively, the bit sequence being used to test the BERT is a sequences of alternating ones and zeros. [3]

Two separate PCBs are used for transmitting and receiving the bitstreams. This BERT can test for data rates up to 370kbps, which is signicantly lower than this project's targetted data rates. As a result of using a periodic square wave, the issue of synchronising the transmitted and received bitstreams is simplified. This system has significantly lower cost and complexity in comparison to FPGA equivalents. Unfortunately, in reality, bitstreams won't be sequential ones and zeros and would appear more random, meaning the displayed error rate may not necessarily reflect real world conditions for data transmission. [3]

2.3 Microcontroller-based BER

An STM32 Microcontroller combined with a Complex Programmable Logic Device (CPLD) was used to make a BER tester. A specific type of communication channel is not specified. This system uses pseudorandom patterns to test a channel. [4]

A single STM32 is used for both transmitting and receiving the bitstreams and tests the error rate at 2.048Mbps. [4] This data rate is similar to the data rate this project is targetting. Due to using only one device, this may not accurately simulate real world conditions.

Overall, both FPGAs and Microcontrollers seem as valid options in building a BER tester since they can both be programmed to transmit and receive a PRBS and compare bitstreams. However, in order to best simulate real world data transmission, two separate devices for transmitting and receiving would be ideal. While this may increase cost, relative to the price of BER equipment, it shouldn't be significant. [2, 3]

3 Work Performed to-date

3.1 Choice of Hardware

In this project, there is choice of hardware between using Microcontrollers or FPGAs. To detemine which is suitable, the capabilities of a device of each were tested using a 10MHz clock signal, which would allow for data rate of 10Mbps.

3.1.1 MC

The board being tested here is the NUCLEO-F401RE board which has an STM32F401RET6 Microcontroller using the Serial Peripheral Interface (SPI) synchronous serial communication protocol. [6, 7]

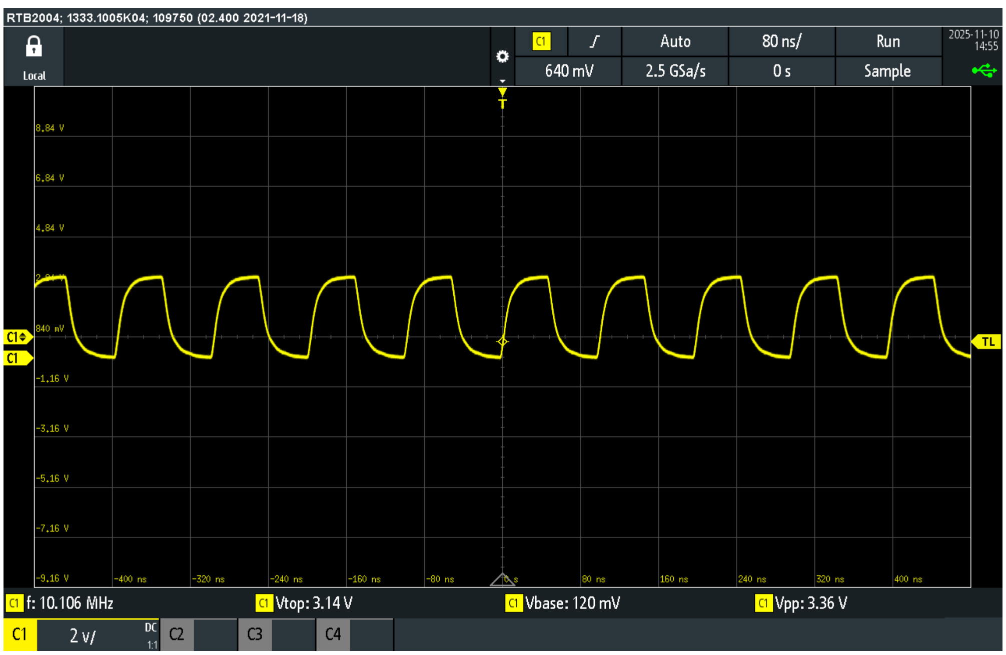

Figure 1: Oscilloscope capture of a 10MHz clock from the NUCLEO-F401RE

In Figure 1, the clock signal is the correct frequency with an adequate peak-to-peak voltage. However, the signal appears to rise and fall exponentially rather than instantaneously like a typical square wave would. Different frequencies of clock signals can be used to test for this effect.

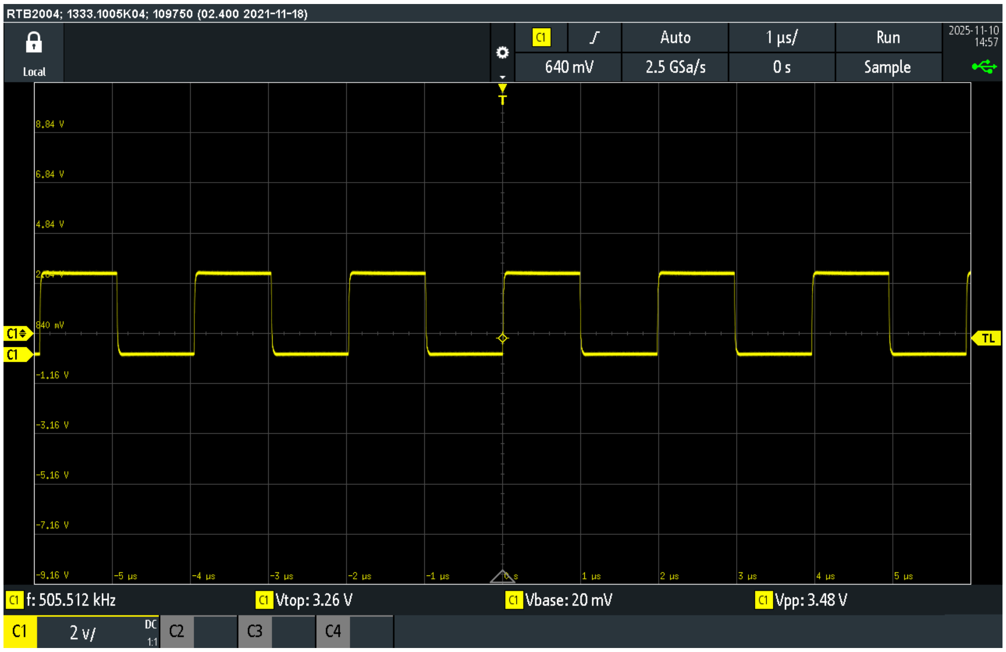

Figure 2: Oscilloscope capture of a 500kHz clock from the NUCLEO-F401RE

In Figure 2, it can be seen that the microcontroller produces a conventional square wave clock signal at 500kHz, suggesting that there is a frequency limitation in the measurement equipment.

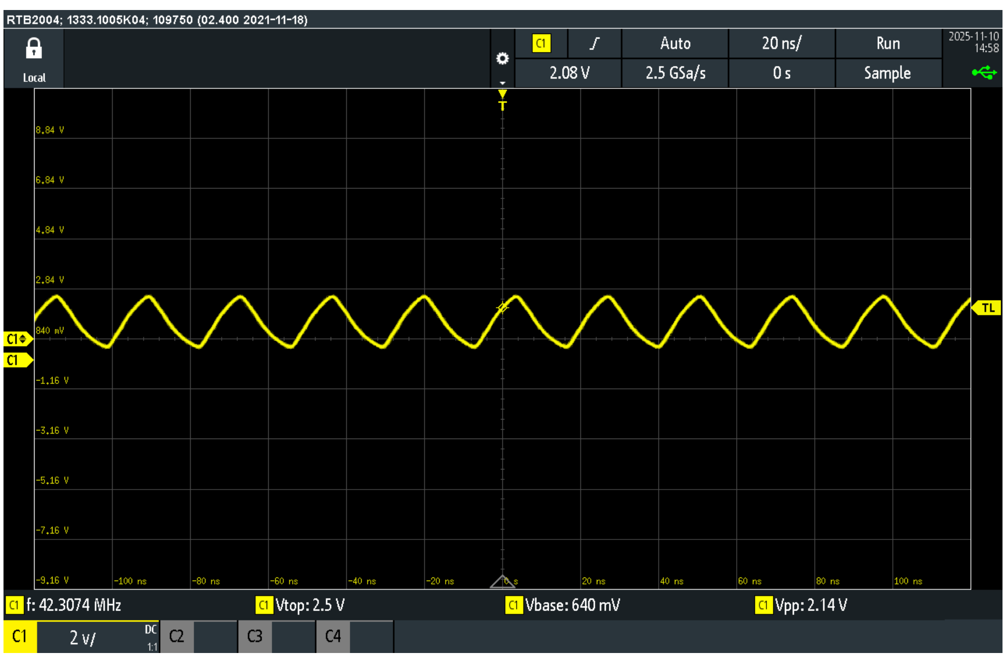

Figure 3: Oscilloscope capture of a 42MHz clock from the NUCLEO-F401RE

In Figure 3, the microcontroller is set to output a clock signal at its highest possible frequency of 42MHz. [6, 7] The signal is almost triangular, but still holds a fundamental frequency of 42MHz. However, the peak-to-peak voltage (2.14V) is attenuated compared to that at 10MHz (3.36V) in Figure 1 and 500kHz (3.48V) in Figure 2.

The oscilloscope probe being used has a -3dB bandwidth of 10MHz which explains the distortion in the 10MHz and 42MHz clock signals. This is because in the Fourier Series of a square wave, along with its fundamental frequency, there are other higher frequencies that allow the wave to be square. [8] So even in a 10MHz square wave, which is exactly at the cut off frequency, there are high frequency harmonics that exist which allow for the wave to be square. These frequencies are attenuated, giving the output in Figure 1. But since the 10MHz frequency itself is within the bandwidth, there is little to no attenuation in the peak-to-peak voltage, like there is at 42MHz in Figure 2.

3.1.2 FPGA

4 Project Management

5 Future Work

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